Assignment 5: Database Creation and Table Implementation

Hospital Management System - SQL DDL implementation of the finalized schema

1. Objective

This assignment creates the database for the Hospital Management System and implements all ten tables from the finalized relational schema using SQL DDL. The tables are created in **dependency order** (parents before children) so that every foreign key can reference an existing table.

2. Environment

The script uses standard SQL types and syntax and is written to run on **MySQL 8.0** (CHECK constraints are enforced from 8.0.16). All outputs shown in this report were produced by executing the same statements on SQLite 3.45 with foreign-key enforcement enabled (`PRAGMA foreign_keys = ON`), so the verification output uses SQLite's catalog commands. In MySQL the equivalents are `SHOW TABLES` and `DESCRIBE table_name`.

3. Creating the Database

```
CREATE DATABASE hospital_management;
USE hospital_management;
-- SQLite: no CREATE DATABASE is needed; the database is the file opened (sqlite3 hospital.db)
-- and foreign keys are switched on per connection with:  PRAGMA foreign_keys = ON;
```

4. Table Creation (SQL DDL)

Design decisions: primary keys on every table; a composite primary key on PRESCRIPTION_MEDICINE; foreign keys on the many side of every 1:M relationship; **ON DELETE RESTRICT** on clinical/financial parents so history cannot be orphaned or lost, and **ON DELETE CASCADE** only where line items have no meaning without their parent (PRESCRIPTION_MEDICINE). BILL.admission_id is nullable because outpatient bills have no admission. The meaning of each constraint is explained in Assignment 6.

```
CREATE TABLE DEPARTMENTS (
    department_id    INT          PRIMARY KEY,
    department_name  VARCHAR(50)  NOT NULL UNIQUE,
    location         VARCHAR(60)  NOT NULL
);

CREATE TABLE PATIENTS (
    patient_id  INT          PRIMARY KEY,
    name        VARCHAR(60)  NOT NULL,
    DOB         DATE         NOT NULL CHECK (DOB >= '1900-01-01'),
    Gender      VARCHAR(10)  NOT NULL CHECK (Gender IN ('Male','Female','Other')),
    ph_no       VARCHAR(15)  NOT NULL CHECK (LENGTH(ph_no) >= 10),
    address     VARCHAR(120),
    CONSTRAINT uq_patient_identity UNIQUE (name, DOB, ph_no)
);

CREATE TABLE DOCTOR (
    doctor_id       INT          PRIMARY KEY,
    doctor_name     VARCHAR(60)  NOT NULL,
    specialization  VARCHAR(50)  NOT NULL,
    ph_no           VARCHAR(15)  NOT NULL UNIQUE,
    email           VARCHAR(80)  NOT NULL UNIQUE CHECK (email LIKE '%_@_%.%'),
    department_id   INT          NOT NULL,
    CONSTRAINT fk_doctor_dept FOREIGN KEY (department_id)
        REFERENCES DEPARTMENTS(department_id) ON UPDATE CASCADE ON DELETE RESTRICT
);

CREATE TABLE APPOINTMENT (
    appointment_id      INT          PRIMARY KEY,
    appointment_date    DATE         NOT NULL,
    appointment_status  VARCHAR(15)  NOT NULL DEFAULT 'Scheduled'
        CHECK (appointment_status IN ('Scheduled','Completed','Cancelled')),
    patient_id          INT          NOT NULL,
```

```
    doctor_id           INT          NOT NULL,
    CONSTRAINT fk_appt_patient FOREIGN KEY (patient_id)
        REFERENCES PATIENTS(patient_id) ON UPDATE CASCADE ON DELETE RESTRICT,
    CONSTRAINT fk_appt_doctor FOREIGN KEY (doctor_id)
        REFERENCES DOCTOR(doctor_id) ON UPDATE CASCADE ON DELETE RESTRICT
);

CREATE TABLE ADMISSIONS (
    admission_id    INT          PRIMARY KEY,
    admission_date  DATE         NOT NULL,
    discharge_date  DATE,
    room_number     VARCHAR(10)  NOT NULL,
    patient_id      INT          NOT NULL,
    doctor_id       INT          NOT NULL,
    CONSTRAINT chk_discharge CHECK (discharge_date IS NULL OR discharge_date >= admission_date),
    CONSTRAINT fk_adm_patient FOREIGN KEY (patient_id)
        REFERENCES PATIENTS(patient_id) ON UPDATE CASCADE ON DELETE RESTRICT,
    CONSTRAINT fk_adm_doctor FOREIGN KEY (doctor_id)
        REFERENCES DOCTOR(doctor_id) ON UPDATE CASCADE ON DELETE RESTRICT
);

CREATE TABLE MEDICAL_RECORD (
    record_id    INT           PRIMARY KEY,
    diagnosis    VARCHAR(100)  NOT NULL,
    treatment    VARCHAR(200),
    record_date  DATE          NOT NULL DEFAULT (CURRENT_DATE),
    patient_id   INT           NOT NULL,
    doctor_id    INT           NOT NULL,
    CONSTRAINT fk_rec_patient FOREIGN KEY (patient_id)
        REFERENCES PATIENTS(patient_id) ON UPDATE CASCADE ON DELETE RESTRICT,
    CONSTRAINT fk_rec_doctor FOREIGN KEY (doctor_id)
        REFERENCES DOCTOR(doctor_id) ON UPDATE CASCADE ON DELETE RESTRICT
);

CREATE TABLE BILL (
    bill_id         INT            PRIMARY KEY,
    total_amount    DECIMAL(10,2)  NOT NULL CHECK (total_amount >= 0),
    payment_status  VARCHAR(10)    NOT NULL DEFAULT 'Pending'
        CHECK (payment_status IN ('Pending','Partial','Paid')),
    bill_date       DATE           NOT NULL DEFAULT (CURRENT_DATE),
    patient_id      INT            NOT NULL,
    admission_id    INT,           -- NULL for outpatient (OPD) bills
    CONSTRAINT fk_bill_patient FOREIGN KEY (patient_id)
        REFERENCES PATIENTS(patient_id) ON UPDATE CASCADE ON DELETE RESTRICT,
    CONSTRAINT fk_bill_adm FOREIGN KEY (admission_id)
        REFERENCES ADMISSIONS(admission_id) ON UPDATE CASCADE ON DELETE RESTRICT
);

CREATE TABLE PRESCRIPTION (
    prescription_id    INT   PRIMARY KEY,
    prescription_date  DATE  NOT NULL DEFAULT (CURRENT_DATE),
    patient_id         INT   NOT NULL,
    doctor_id          INT   NOT NULL,
    CONSTRAINT fk_rx_patient FOREIGN KEY (patient_id)
        REFERENCES PATIENTS(patient_id) ON UPDATE CASCADE ON DELETE RESTRICT,
    CONSTRAINT fk_rx_doctor FOREIGN KEY (doctor_id)
        REFERENCES DOCTOR(doctor_id) ON UPDATE CASCADE ON DELETE RESTRICT
);

CREATE TABLE MEDICINES (
    medicine_id     INT           PRIMARY KEY,
    medicine_name   VARCHAR(60)   NOT NULL,
    manufacturer    VARCHAR(60)   NOT NULL,
    price           DECIMAL(8,2)  NOT NULL CHECK (price > 0),
    stock_quantity  INT           NOT NULL DEFAULT 0 CHECK (stock_quantity >= 0),
    CONSTRAINT uq_medicine UNIQUE (medicine_name, manufacturer)
);

CREATE TABLE PRESCRIPTION_MEDICINE (
    prescription_id  INT  NOT NULL,
    medicine_id      INT  NOT NULL,
    quantity         INT  NOT NULL DEFAULT 1 CHECK (quantity > 0),
    PRIMARY KEY (prescription_id, medicine_id),
    CONSTRAINT fk_pm_rx FOREIGN KEY (prescription_id)
        REFERENCES PRESCRIPTION(prescription_id) ON UPDATE CASCADE ON DELETE CASCADE,
    CONSTRAINT fk_pm_med FOREIGN KEY (medicine_id)
        REFERENCES MEDICINES(medicine_id) ON UPDATE CASCADE ON DELETE RESTRICT
);
```

5. Verification of the Implementation

5.1 Tables created

```
SELECT name AS table_name FROM sqlite_master WHERE type = 'table' ORDER BY name;   -- MySQL: SHOW TABLES;
```

table_name
ADMISSIONS
APPOINTMENT
BILL
DEPARTMENTS
DOCTOR
MEDICAL_RECORD
MEDICINES
PATIENTS
PRESCRIPTION
PRESCRIPTION_MEDICINE

10 row(s) returned

5.2 Table structures

Structure of each table (SQLite: `PRAGMA table_info`; MySQL: `DESCRIBE`). notnull = 1 means NOT NULL, pk > 0 marks primary-key columns (1 and 2 for the composite key).

DEPARTMENTS

column_name	data_type	notnull	default_value	pk
department_id	INT	0	NULL	1
department_name	VARCHAR(50)	1	NULL	0
location	VARCHAR(60)	1	NULL	0

3 row(s) returned

PATIENTS

column_name	data_type	notnull	default_value	pk
patient_id	INT	0	NULL	1
name	VARCHAR(60)	1	NULL	0
DOB	DATE	1	NULL	0
Gender	VARCHAR(10)	1	NULL	0
ph_no	VARCHAR(15)	1	NULL	0
address	VARCHAR(120)	0	NULL	0

6 row(s) returned

DOCTOR

column_name	data_type	notnull	default_value	pk
doctor_id	INT	0	NULL	1
doctor_name	VARCHAR(60)	1	NULL	0
specialization	VARCHAR(50)	1	NULL	0
ph_no	VARCHAR(15)	1	NULL	0
email	VARCHAR(80)	1	NULL	0
department_id	INT	1	NULL	0

6 row(s) returned

APPOINTMENT

column_name	data_type	notnull	default_value	pk
appointment_id	INT	0	NULL	1
appointment_date	DATE	1	NULL	0
appointment_status	VARCHAR(15)	1	'Scheduled'	0
patient_id	INT	1	NULL	0
doctor_id	INT	1	NULL	0

5 row(s) returned

ADMISSIONS

column_name	data_type	notnull	default_value	pk
admission_id	INT	0	NULL	1
admission_date	DATE	1	NULL	0
discharge_date	DATE	0	NULL	0
room_number	VARCHAR(10)	1	NULL	0
patient_id	INT	1	NULL	0
doctor_id	INT	1	NULL	0

6 row(s) returned

MEDICAL_RECORD

column_name	data_type	notnull	default_value	pk
record_id	INT	0	NULL	1
diagnosis	VARCHAR(100)	1	NULL	0
treatment	VARCHAR(200)	0	NULL	0
record_date	DATE	1	CURRENT_DATE	0
patient_id	INT	1	NULL	0
doctor_id	INT	1	NULL	0

6 row(s) returned

BILL

column_name	data_type	notnull	default_value	pk
bill_id	INT	0	NULL	1
total_amount	DECIMAL(10,2)	1	NULL	0
payment_status	VARCHAR(10)	1	'Pending'	0
bill_date	DATE	1	CURRENT_DATE	0
patient_id	INT	1	NULL	0
admission_id	INT	0	NULL	0

6 row(s) returned

PRESCRIPTION

column_name	data_type	notnull	default_value	pk
prescription_id	INT	0	NULL	1
prescription_date	DATE	1	CURRENT_DATE	0
patient_id	INT	1	NULL	0
doctor_id	INT	1	NULL	0

4 row(s) returned

MEDICINES

column_name	data_type	notnull	default_value	pk
medicine_id	INT	0	NULL	1
medicine_name	VARCHAR(60)	1	NULL	0
manufacturer	VARCHAR(60)	1	NULL	0
price	DECIMAL(8,2)	1	NULL	0
stock_quantity	INT	1	0	0

5 row(s) returned

PRESCRIPTION_MEDICINE

column_name	data_type	notnull	default_value	pk
prescription_id	INT	1	NULL	1
medicine_id	INT	1	NULL	2
quantity	INT	1	1	0

3 row(s) returned

5.3 Relationships (foreign keys)

```
SELECT m.name AS child_table, f."from" AS fk_column, f."table" AS parent_table, f."to" AS parent_key,
       f.on_delete, f.on_update
FROM sqlite_master m, pragma_foreign_key_list(m.name) f
WHERE m.type = 'table'
ORDER BY m.name;      -- MySQL: SELECT ... FROM information_schema.KEY_COLUMN_USAGE
```

child_table	fk_column	parent_table	parent_key	on_delete	on_update
ADMISSIONS	doctor_id	DOCTOR	doctor_id	RESTRICT	CASCADE
ADMISSIONS	patient_id	PATIENTS	patient_id	RESTRICT	CASCADE
APPOINTMENT	doctor_id	DOCTOR	doctor_id	RESTRICT	CASCADE
APPOINTMENT	patient_id	PATIENTS	patient_id	RESTRICT	CASCADE
BILL	admission_id	ADMISSIONS	admission_id	RESTRICT	CASCADE
BILL	patient_id	PATIENTS	patient_id	RESTRICT	CASCADE
DOCTOR	department_id	DEPARTMENTS	department_id	RESTRICT	CASCADE
MEDICAL_RECORD	doctor_id	DOCTOR	doctor_id	RESTRICT	CASCADE
MEDICAL_RECORD	patient_id	PATIENTS	patient_id	RESTRICT	CASCADE
PRESCRIPTION	doctor_id	DOCTOR	doctor_id	RESTRICT	CASCADE
PRESCRIPTION	patient_id	PATIENTS	patient_id	RESTRICT	CASCADE
PRESCRIPTION_MEDICINE	medicine_id	MEDICINES	medicine_id	RESTRICT	CASCADE
PRESCRIPTION_MEDICINE	prescription_id	PRESCRIPTION	prescription_id	CASCADE	CASCADE

13 row(s) returned

5.4 Relationship summary

Parent (1)	Child (M)	Relationship	Foreign key
PATIENTS	APPOINTMENT	Has	APPOINTMENT.patient_id
DOCTOR	APPOINTMENT	Assigned to	APPOINTMENT.doctor_id
PATIENTS	ADMISSIONS	Gets admission	ADMISSIONS.patient_id
DOCTOR	ADMISSIONS	Handles	ADMISSIONS.doctor_id
DEPARTMENTS	DOCTOR	Works in	DOCTOR.department_id
PATIENTS	MEDICAL_RECORD	Has	MEDICAL_RECORD.patient_id
DOCTOR	MEDICAL_RECORD	Creates	MEDICAL_RECORD.doctor_id
PATIENTS	PRESCRIPTION	Has	PRESCRIPTION.patient_id
DOCTOR	PRESCRIPTION	Prescribes	PRESCRIPTION.doctor_id
PATIENTS	BILL	Pays	BILL.patient_id
ADMISSIONS	BILL	Generates	BILL.admission_id
PRESCRIPTION / MEDICINES	PRESCRIPTION_MEDICINE	Includes (M:N resolved)	prescription_id, medicine_id (composite PK)

Note on the design document: *Section 4 of the design document also lists ADMISSIONS-MEDICAL_RECORD and MEDICAL_RECORD-PRESCRIPTION as 1:M relationships, but the finalized relational schema (Section 5) has no foreign key for them. This implementation follows the finalized schema exactly; adding those two foreign keys is proposed in Assignment 3 (Section 5.2).*